

Jayadratha Gayen

jayadratha.gayen@research.iiit.ac.in | LinkedIn | Github | Google Scholar | Homepage

EDUCATION

International Institute of Information Technology, Hyderabad

M.Sc in Electronics and Communication Engineering by Research (CGPA: 8.29/10)

Hyderabad, India

July 2022 - July 2025

Indian Institute of Engineering Science and Technology, Shibpur

Bachelor of Technology in Electrical Engineering (CGPA: 8.02/10)

Howrah, India

July 2014 - June 2018

RESEARCH EXPERIENCE

Research Fellow

Machine Learning Lab - MLL (IIITH)

Nov 2022 - July 2025

Hyderabad, India

- worked in graph neural network (GNNs), uncertainty management, 3D vision
- Conducted research on GNNs, high-stakes applications like legal judgment prediction, disease prediction; accepted at TMLR 2025, DAI AAAI 2025
- Explored continuous time dynamic graphs (CTDGs), reject option classification; submitted in ASONAM 2025, TGL KDD 2025
- Worked on 3D computer vision application for pose transfer and motion transfer

PUBLICATIONS

1. Reducing Misclassification Risk in Dynamic Graph Neural Networks through Abstention

Accepted at ASONAM 2025

Jayadratha Gayen, Himanshu Pal, Naresh Manwani, Charu Sharma

[[ArXiv](#), [Code](#)]

2. Confidence First: Reliability-Driven Temporal Graph Neural Networks

Accepted at KDD 2025 Temporal Graph Learning (TGL) Workshop

Jayadratha Gayen, Himanshu Pal, Naresh Manwani, Charu Sharma

[[Openreview](#), [ArXiv](#), [Code](#)]

3. Node Classification With Reject Option

Accepted at TMLR 2025

Uday Bhaskar, Jayadratha Gayen, Charu Sharma, Naresh Manwani

[[Openreview](#), [ArXiv](#)]

4. Node Classification With Integrated Reject Option For Legal Judgement Prediction

Accepted at AAAI 2025 Deployable AI (DAI) Workshop

Uday Bhaskar, Jayadratha Gayen, Charu Sharma, Naresh Manwani

[[ArXiv](#), [Slides](#)]

WORK EXPERIENCE

Open Source Developer

Google Summer of Code at INCF

May 2025 - Sep 2025

Remote

- **Project:** DevoTG: Dynamic Graph Neural Networks for Modeling C. elegans Development.
- Model C. elegans embryogenesis (cell division) and connectome formation using temporal graph methods.
- Modeled spatiotemporal cell division data and neuronal network formation as evolving graphs.
- Implemented TGN-based models for representation learning and prediction of developmental patterns.
- Built interactive 3D visualizations and dynamic network plots for biological data exploration.

Engineer

Vikram Solar Limited

July 2018 - Dec 2019

Falta, West Bengal, India

- Contributed to a Six Sigma project on rolled throughput improvement, increasing production by 10%.
- Involved in bottleneck analysis, DOE in Minitab, and Pareto, Five Why, and Fishbone analyses to optimize production quality.

- Analyzed technical data to decide current binning for all module types and introduced RFID tag automation, reducing manual errors. Utilized SAP and extensive Excel analysis for data management and insights.
- Led root cause analysis and KAIZEN projects, achieving a 2% reduction in power consumption.

SELECTED PROJECTS

Nyay Saheli: Conversational RAG for Indian Women

2025

[\[GitHub\]](#)

- Developed a conversational AI assistant using Retrieval-Augmented Generation (RAG) to provide empathetic legal guidance to Indian women in multiple Indian languages.
- Implemented multilingual natural language processing capabilities to deliver culturally sensitive and legally accurate information for women's rights and safety.
- Designed the system to handle complex legal queries while maintaining empathetic tone and cultural context appropriate for diverse Indian communities.

Node Classification With Integrated Reject Option

2024

[\[Paper Link\]](#)

- Introduced a novel graph neural network framework, NodeCwR, incorporating cost-based and coverage-based rejection strategies to enhance the reliability of Legal Judgment Prediction (LJP) and disease prediction tasks.
- Achieved a 6-19% improvement in prediction accuracy on the ILDC dataset for LJP and abstaining from uncertain cases.
- Enhanced interpretability of the rejection decisions using SHAP visualizations.

Predict Confidently, Predict Right: Abstention in Dynamic Graph Learning

2024

[\[Paper Link\]](#)

- Developed a novel reject option strategy for CTDGs, allowing the model to abstain from uncertain predictions, enhancing reliability in link prediction and node classification tasks
- Achieved 10-15% improvement in AUC/AP scores across six dynamic graph datasets, effectively addressing class imbalance and critical misclassification
- Demonstrated the model's effectiveness through exhaustive experiments in dynamic and risk-sensitive domains such as fraud detection and disease prediction.

3D Motion Transfer Across Diverse Character Topologies

2023

[\[Project Output\]](#)

- Developed a novel method using Graph Convolution Networks (GCN) for realistic 3D motion transfer between diverse character topologies, including humanoids and quadrupeds.
- By incorporating prior frame encodings, the approach reduces jitter and ensures smoother, natural animations.
- The method outperforms traditional pose transfer techniques, especially on complex characters, as shown on the Mixamo dataset.

Multilingual News Article Similarity Detection Using Siamese Networks

2022

[\[Project page\]](#)

- Developed a multilingual news article similarity detection system using a Siamese architecture.
- Three models were implemented: TF-IDF with MLP, GloVe with LSTM, and multilingual DistilBERT, achieving the best result with DistilBERT (PCC: 0.5683).
- The system excels in cross-linguistic news comparison, outperforming baseline models in SemEval 2022 Task 8.

TEACHING EXPERIENCE

Teaching Assistant

May 2023

3D Vision Summer School

CVIT, IIT Hyderabad, India

- Conducted a tutorial on graph neural networks, their current progress, and case studies on the usage of GNNs for 3D vision especially mesh-based systems.
- Taken doubt-solving sessions in GNNs and 3D vision concepts for the participants.

Instructor

October 2023

Executive Training Program on AIML

ihub-data, IIT Hyderabad, India

- Coordinated in two-weeks executive training program on machine learning and artificial intelligence.
- conducted tutorials concepts like PCA, end-to-end ML model building. Also taken doubt-solving sessions.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

Developer Tools: GitHub, VS Code, Conda, Poetry

Libraries: PyTorch, PyG, scikit-learn, TensorFlow, Keras, Transformers

Softwares: Blender, MeshLab, Minitab, MATLAB Simulink

RELEVANT COURSES

Topics in Deep Learning, Computer Vision, Statistical Methods in AI, Applied Optimization, Linear Algebra, Calculus, Probability & Statistics, Digital Signal Processing

EXTRACURRICULAR & PARTICIPATION

- **Founding Member, Srijani - The Hope:** Started an independent non-profit NGO focused on improving education and healthcare in remote areas of West Bengal, India, with limited accessibility. (*Mar 2020 - Apr 2022*)
- Participated in LoG Delhi Meetup, November 26, 2024
- Attended 8th Summer School on AI, IIIT Hyderabad, July 2024
- National Finalist (top 6/275+ teams), BharatGen Hackathon: Designed *Nyay Saheli*, a GenAI-powered multilingual chatbot offering personalized legal guidance for Indian women. (*Apr-June 2025*)

REFERENCES

Below are individuals who can vouch for my skills, expertise, and contributions:

- **Prof. Charu Sharma (*Preferred*)** - Assistant Professor, Machine Learning Lab, IIIT Hyderabad - charu.sharma@iiit.ac.in
- **Prof. Anindita Sengupta** - Head, Electrical Engineering, IEST Shibpur - hod@ee.iests.ac.in
- **Dr. Bradly Alicea** - Head Scientist, Orthogonal Research and Education Laboratory; Lecturer, iSchool @ UIUC - balicea@openworm.org
- **Prof. Naresh Manwani** - Assistant Professor, Machine Learning Lab, IIIT Hyderabad - naresh.manwani@iiit.ac.in